

Tan Xin Er

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EDUCATION

Bachelor of Engineering in ICT majoring in Software Engineering (Honours with Merit) | May 2026

Singapore Institute of Technology (SIT), Singapore

Diploma in Accountancy | March 2022

Singapore Polytechnic (SP), Singapore

WORK EXPERIENCE

IT Application Intern

SBS Transit Pte Ltd

Singapore

May 2025 – April 2026

- Developed and enhanced internal enterprise applications using .NET (C#, ASP.NET), JavaScript, SQL, and REST APIs, implementing backend business logic, stored procedures, database integrations, and frontend functionality to support end-to-end application workflows and improve system reliability.
- Created user-facing features including data entry forms, validation logic, and interactive interfaces, improving usability and enabling efficient user interactions across enterprise applications.

SmartTDG (Capstone) | [Project Link](#)

- Delivered an AI-powered technical documentation application using React, FastAPI, Python, LangChain, Ollama, and local LLMs, providing IT teams with an interactive platform to analyze legacy .NET applications and automate technical documentation workflows, reducing manual analysis time by over 95%.
- Designed and implemented full-stack application workflows by developing React-based user interfaces and FastAPI backend services to support code uploads, processing status tracking, API communication, and automated documentation generation.
- Implemented a multi-language static analysis pipeline using Roslyn Analyzer and custom parsers for C#, VB.NET, JavaScript, HTML, and SQL to extract code structures, dependencies, relationships, and database interactions.
- Engineered scalable backend processing pipelines to handle large legacy codebases through optimized parsing, chunking strategies, asynchronous workflows, and prompt engineering techniques to improve AI context handling and output quality.

Data Engineer Intern

Munich Re

Singapore

December 2024 – April 2025

- Reduced data processing time by over 50% by developing automated R-based ETL workflows, streamlining data integration, transformation, and reporting processes.
- Enhanced and maintained existing R scripts by refactoring processing logic, improving code readability, and strengthening maintainability for ongoing analytical workflows.

Junior Research Assistant Intern

Illinois Advanced Research Centre

Singapore

May 2024 – July 2024

- Developed a full-stack network monitoring dashboard using React, Flask, Python, and MongoDB to visualize industrial control system traffic and security insights.
- Built REST APIs for authentication, CVE management, device tracking, and data processing while integrating dynamic vulnerability records.
- Developed data pipelines for network analysis, vulnerability mapping, and dynamic security intelligence storage.
- Integrated Zeek, Prometheus, and Grafana for real-time network monitoring, metrics collection, and security visualization.

PROJECTS / EXPERIENCES

Booking Platform | JavaScript, HTML, CSS, Node.js, Express.js, REST APIs, MySQL, MongoDB | [Project Link](#)

- Engineered a full-stack hotel and restaurant booking platform using Node.js, Express.js, JavaScript, MySQL, and MongoDB, enabling users to discover, book, and manage accommodations and dining reservations.
- Designed a hybrid MySQL–MongoDB database architecture to manage transactional, geospatial, and dynamic user profile data, leveraging MongoDB geospatial indexing to implement location-based restaurant recommendations.
- Built RESTful APIs for bookings, reviews, search, and user profiles, with SQL joins, aggregations, and window functions for analytics.

Semantic Segmentation Mobile App (Autonomous Driving) | Python, Flutter, Dart, AI/ML | [Project Link](#)

- Developed a Flutter/Dart mobile computer vision application integrating AI-based semantic segmentation models for real-time video and image inference on edge devices.
- Implemented live camera stream and single-image inference workflows, optimizing model execution for responsive on-device scene understanding and mobile user experiences.
- Built Python-based model training pipelines for BiSeNet and MobileNet-UNet architectures, using TensorFlow conversion workflows to optimize trained models for mobile deployment.

Real-Time Multi-Modal Assistive Intelligence System | Python, Flask, JavaScript, React, MQTT | [Project Link](#)

- Engineered a full-stack edge computing accessibility platform using React, Flask, Python, and JavaScript, integrating real-time AI-powered ASL recognition, speech-to-text, text-to-speech, and facial emotion recognition.
- Designed a low-latency communication architecture using MQTT, Socket.IO, and REST APIs to stream data between Raspberry Pi edge devices and a web-based dashboard.
- Implemented responsive dashboard interfaces to visualize real-time recognition results, transcriptions, and emotion analytics with seamless frontend-backend integration.

Secure Backend System with AWS IAM Authentication | C#, SQL, AWS IAM, REST APIs

- Executed a REST API to support backend data retrieval and business workflows, implementing SQL based data handling and validation logic.
- Secured API endpoints using AWS IAM (users, roles, and policies) and built a C# console application to consume and validate authenticated endpoints end-to-end.

GitLab CI/CD Pipeline Automation | GitLab, Git, CI/CD, Deployment Automation

- Designed and implemented a GitLab CI/CD pipeline to automate software build, testing, and deployment workflows across development environments and production releases, improving development efficiency, deployment consistency, and overall software delivery reliability.

SKILLS

Python, C#, JavaScript, SQL, C, React, Flask, .NET, Node.js, REST APIs, MySQL, MongoDB, Docker, AWS, Git

AWARDS / ACHIEVEMENTS

EPS Computer Systems Scholarship (AY2023/2024 – AY2025/2026)

- Awarded the scholarship for three consecutive years

SBS Hackathon 2025 - 1st Runner Up

- Created an ESP8266 LED control system using MQTT for real-time remote brightness adjustment, optimizing power consumption and display visibility.

CERTIFICATIONS

- SCRUM Master
- Microsoft Office Excel & Excel Expert

LANGUAGE

English: Fluent

Chinese: Intermediate

Bahasa Indonesia: Basic